Contents

RESEARCH FINDINGS — Conflux Atlas, 14 July 2026

Scope: everything measured between Phase 0 (schema gates) and Bridge v2 (nugget-calibrated backfill), one build day. **Posture:** objective; misses reported with the same precision as wins. Every number below is reproducible from a committed artifact (`data-validation-reports/`) and a runner script. **Suite:** 235 tests green. Branches merged: PRs #1–#5.

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1. System under study

A confidence-aware evidence system over religious-demography share series for MENA polities + diaspora. State is a set of *anchors* and *observations*: source `s` states share `x ∈ [0,1]` for (polity `p`, group `g`, year `t`) with confidence `c`. Around this sit five measured subsystems:

1. **Trust ledger** — Beta-Bernoulli posteriors per hypothesis, settlement-only updates.
2. **Connascence layer** — typed edges (structural / conceptual / co-variance / temporal) that change how settlements are *weighted and routed*, not what is claimed.
3. **LLM proposer** — a scored, verifier-gated enrichment agent inside the same ledger.
4. **Banded forecaster + pre-registered backtest** — share forecasts as central intervals, Winkler-scored.
5. **Sparsity bridge** — modern-fit dynamics backfilled into sparse eras with calibrated bands.

2. Trust arithmetic (Phases 2–2b)

2.1 Graded Beta-Bernoulli updates

Posterior per hypothesis `h`: $\theta_h \sim \mathrm{Beta}(\alpha, \beta)$, prior $\mathrm{Beta}(1, 1)$. A settlement with outcome $y \in \{0, 1\}$ and weight $w \in (0, 1]$:

$$(\alpha, \beta) \leftarrow (\alpha + w\,y,\ \beta + w\,(1 - y))$$

`w < 1` implements (a) the **independence discount** — corroborations between same-method-family sources (census/survey/synthesis/derived/scholarly registry) carry `w = 0.5`, because two surveys agreeing is weaker evidence than a survey agreeing with a census; and (b) **partial settlement** — one-hop propagation along strict-tier co-variance edges at `w = strength × coefficient`, capped ≤ 0.25.

2.2 Level-scaled corroboration tolerance

A flat ±5 pp tolerance is absurd at trace shares (any two near-zero values "agree"). Tolerance is binned by level: trace ±0.5 pp, minority ±1 pp, significant ±3 pp, else ±5 pp; observations < 0.5 pp are non-reports and never settle.

2.3 Definitional routing — measured effect

377 corroboration claims re-settled with discounts + routing (14 claims rerouted off `source_trust:*` onto `definition_gap:*` lanes):

Hypothesis	Unweighted mean	Weighted + routed	Δ
`jewishdatabank_wjp`	0.667	0.500	**−0.167**
`cbs_population_madaf`	0.667	0.750	**+0.083**
`pew_2010_2020`	0.700	0.688	−0.012
`definition_gap:cjp_vs_pew_jewish`	—	0.800	new lane
`definition_gap:cbs_arab_proxy_vs_pew_muslim`	—	**0.167**	new lane

Finding: WJP's trust was inflated by definition-overlap agreement with Pew (core-Jewish vs affiliation), and CBS was drained by an Arab-proxy-vs-muslim definition mismatch that is now quarantined in its own lane. The 0.167 on that lane is itself a result: the proxy systematically over-counts (folds in Arab Christians and Druze).

3. Co-variance discovery: the multiple-testing result (Phase 2b)

For series pair (A, B), over all overlapping *moving* transition pairs, agreement is directional co-movement. The null is **bucket-stratified**: for transitions with origin place-hashes a, b,

$$p_{\text{null}} = \overline{p_a p_b + (1-p_a)(1-p_b)}, \qquad p_h = P(\text{up} \mid \text{moving},\ \text{hash} = h)$$

so "both series are secularly declining" cannot mint an edge. Admission: $n \geq 3$ overlaps, one-sided exact binomial $P(X \geq k \mid n, p_{\text{null}}) < 0.05$, **and** Benjamini–Hochberg survival across all scored pairs ($p_{(i)} \leq \alpha\, i/m$).

Measured on 3,852 scored cross-polity pairs:

- raw α tier: **4 real edges** (best p = 0.012 at n = 7);
- shuffle control (outcomes permuted within series): **20 raw-α edges** — *more than the real tape*;
- BH-FDR survivors: **0**, real and shuffled.

Finding: at thousands of candidates with n = 3–9 overlaps each, raw α admits noise faster than signal — the shuffle control proved it quantitatively, and the FDR gate is not optional. Consequence wired into code: a **strict tier** (FDR survivors; the only input to partial settlement — currently empty, so partial settlement correctly fired 0 times) and a **hypothesis tier** (raw-α; clusters/REVIEW/LLM input only). The bottleneck is data density (anchor grid 1900/1950/2000/2010/2020 caps overlaps at ~2 per pair), not the statistic.

4. LLM proposer under scorekeeping (Phase 2b → full population)

Protocol: local `qwen3:8b`, deterministic decoding (T=0, top-k 1, seed 42), closed vocabularies, JSON schema-forced, windowed (12 items/call); every proposal enters the ledger as a claim; **deterministic verifiers are the only promotion path**; abstention is free; the proposer has its own posterior.

Run	Windows	Proposals	Verified	Rejected	Posterior mean
Pilot (with decoys)	6	56	50	6	0.879
Full population	40	467	458	9	**0.979**

Timing: 16.95 s/call mean, 677.9 s total, 0 malformed windows. Error taxonomy from the pilot: all 3 planted no-edge decoys were mislabeled `complement` — and all 3 were caught by verifiers; 2 true `conservation` pairs mislabeled (fairly penalized); 1 unproven `definition` proposal rejected by the systematic-offset test. Event attribution abstained on all offered clusters (correct: the closed event list doesn't explain post-1950 growth co-movements).

Finding: a small local model is a reliable *classifier inside closed vocabularies* when every output is verifier-gated and its accuracy is priced in the same ledger as the data sources. Its measured failure mode is ontology confusion between coupling kinds, not hallucination.

5. Pre-registered banded backtest (Phase 3)

5.1 Protocol

Frozen before running: cut 1975; nominal central coverage 0.80 ($\alpha = 0.20$); policies persistence / reversion / AR1; primary metric mean **Winkler interval score**

$$S_\alpha(\ell, u; y) = (u - \ell) + \tfrac{2}{\alpha}(\ell - y)\,\mathbf{1}[y < \ell] + \tfrac{2}{\alpha}(y - u)\,\mathbf{1}[y > u]$$

(lower is better; charges width *and* misses); success rule "candidate beats all baselines on primary metric"; contested sources (McCarthy) excluded from validation targets; targets = realized post-cut anchor years only (no interpolated truth). Baseline width (w0): half $= z_{0.80}\,\hat{\sigma}_{\text{train}}\sqrt{h/10}$, h = years past last train point.

5.2 First run (baselines only)

Policy	n	Coverage (stated 0.80)	Mean IS
persistence	108	0.593	0.321
reversion	108	0.565	0.327
ar1	0 (abstained)	—	—

Bridge pooled holdouts: 37 settled, coverage **0.297**. Both misses were published as results before any diagnosis.

6. Expanded experimentation (E1–E7)

All coverage numbers now carry Wilson 95% score intervals: $\tilde{p} \pm \frac{z}{1+z^2/n}\sqrt{\hat{p}(1-\hat{p})/n + z^2/4n^2}$.

6.1 The coverage miss is train staleness (E1/E2)

Cuts 1950/1975/1990 produce byte-identical anchor tapes (no train data 1950–2000 → the cut moves nothing). When year-2000 anchors enter (cut 2000): coverage 0.542 → **0.861** [0.762, 0.923], IS 0.354 → 0.153. Desk-augmented series (multi-source timeline; hand-seed and contested sources excluded from *targets*) unlock AR1 (n = 12 at cut 1975 via WJP-1970 third points, coverage 0.833).

6.2 Width ablation with leakage-safe selection (E3)

Cross-cut selection would leak (cut-1950/1990 tapes share realized 2005–2020 target rows with the 1975 tape), so selection used a deterministic **series split**: sha1(polity|group) even → selection, odd → confirmatory. Four width models: w0 (above); w1 walk-forward residual quantiles; w2 level-conditional dynamics σ; w3 **conformal inflation** — w0 shape × the smallest λ (grid) achieving 0.80 empirical coverage on *pre-cut* walk-forward folds. Selection (min mean |cov − 0.80|, lower is better): w3 at 0.003, far ahead of w0/w1 (0.161) and w2 (0.200); fitted λ = 2.5 / 3.0 / 1.0 (persistence/reversion/ar1).

Single confirmatory shot, odd half @1975: persistence coverage 0.420 → **0.681** [0.564, 0.779], IS 0.494 → 0.426. Improved and *still a miss* (0.80 outside the interval); reversion under w3 calibrated (0.812). No second shot taken.

6.3 First candidate to pass the frozen rule (E4)

Analog retrieval — origin-context place vectors (level/prior-delta/gap/vol one-hots + raw share and prior rate, L2-normalized; outcome features excluded from the query by construction), cosine top-k = 25 cross-series neighbors from the train-window transition catalog, forecast distribution = neighbor rate/decade quantiles (10/50/90) × horizon. Fixed hyperparameters, no tuning.

Confirmatory half @1975	n	Coverage	Mean IS
analog	64	0.719 [0.599, 0.814]	**0.415**
persistence (w3)	69	0.681	0.426
reversion (w3)	64	0.812	0.482
climatology floor	642	0.838	0.976

Nominal win under the pre-registered rule. Paired on the 64 shared targets: mean IS difference (persistence − analog) = **+0.041**, bootstrap 95% CI **[−0.113, +0.184]** — spans zero. Analog also abstained on 5 easy targets (persistence IS there: 0.031). Both statements stand; neither is softened. The ensemble candidate (weights grid-fit on the selection half degenerated to pure AR1) entered and lost (IS 1.000).

6.4 Shock split, made meaningful (E5)

Window-only event tagging is degenerate on cut→target windows (any event tags every claim: calm 0 / shock 108). With the polity-contact rule (event's `affected_polities` or a triggered migration edge must touch the claim's polity): calm 97 / shock 33, and

Slice	Coverage	Mean IS	Mean width
calm	0.433	0.413	0.137
shock	0.788	0.589	0.551

Finding: shock series *cover better but score worse* — volatile trains produce wide bands that catch outcomes and pay for width. Coverage alone would have inverted the conclusion; this is the concrete argument for a proper scoring rule as primary metric.

6.5 Controls (E7)

- **Climatology floor** (group train-mean, 10–90% train-dispersion band): IS 1.586 vs persistence 0.457 (desk@1975) — all live policies clear the no-skill floor by ~3×.
- **Permutation control** (realized shares shuffled within group × level-bin strata, 200 draws): IS worsens ×1.18 (desk@1975), ×1.37 (desk@2000), ×1.07 (anchors@1975) — bands are series-specific, weakest on the information-poor anchor tape, as expected.

7. Sparsity→simulation bridge: inversion and repair

7.1 The inverted curve (E6b)

282 leave-one-out holdouts over 37 dense desk series, coverage vs distance-to-nearest-anchor:

Gap	n	Coverage (linear width, no nugget)
1–5 y	210	**0.495** [0.428, 0.562]
6–10 y	15	0.667
11–25 y	6	0.833
26–50 y	51	**0.922** [0.815, 0.969]

Finding: the failure mode is the opposite of the design assumption. At long gaps the rate-uncertainty term $z\sigma \cdot (g/10)$ is honest-to-generous; at short gaps it collapses to zero and the residual error is **cross-source definitional scatter**, priced at nothing.

7.2 The nugget (v2)

Measurement noise estimated from cross-source same-polity-year(±3) spreads: for independent noise, $\text{Var}(x_1 - x_2) = 2\sigma_m^2 \Rightarrow \hat{\sigma}_m = \text{sd}(d)/\sqrt{2}$.

Group	$\hat{\sigma}_m$ (share points)	n pairs
muslim	0.0382	50
christian	0.0458	13
jewish	0.0152	14

Scale check: modern rate volatility is 0.027/decade, so cross-source definitional noise ≈ **1.4 decades of real movement**. Width becomes quadrature: half $= z\sqrt{\sigma_m^2 + (\sigma_{\text{eff}}\, g_*)^2}$, $g_* \in \{g/10, \sqrt{g/10}\}$, with $\sigma_{\text{eff}} = 2\sigma$ over event windows touching the polity (pre-declared multiplier). Nugget applies off-anchor only (anchor exactness preserved; all prior contracts pass unchanged).

7.3 The curve, un-inverted

Gap	Phase 3	v2 (sqrt + nugget)
1–5 y	0.495	**0.848** [0.793, 0.890]
6–10 y	0.667	0.800
26–50 y	0.922	0.882
overall	0.589	**0.851**

Every bucket's Wilson interval contains the stated 0.80; mean bucket miscalibration 0.148 → 0.041; short-gap IS *improved* (0.218 → 0.166), i.e. the wider bands pay for themselves under the proper scoring rule. Product shipped: 4,592 banded rows on a 5-year grid (`BRIDGE_V2_BACKFILL.jsonl`); era-mean widths pre-1920 0.087, 1920–1999 **0.116** (the anchor desert), 2000+ 0.055.

8. Negative and null results (kept on the record)

1. **PortalGC geometry ≠ evidential quality** (Phase 2.5): Lorenz-attractor KEEP/EVICT classification showed no useful correlation with independently learned source trust on this graph.

2. **BH-FDR: 0 strict co-variance edges** — the 1948–72 Jewish positive control is floored out by MIN_SHARE + grid coarseness; partial settlement therefore fired 0 times, by arithmetic.
3. **Conservation claims: 0 settleable** — bracketing 1948–51 edges needs mid-century anchors that don't exist yet (nearest: 1900/1950).
4. **AR1 silence on anchor tapes** — honest abstention (≥3 pre-cut points required), not a bug.
5. **anchors@2010 coverage 0.144 is a degeneracy**: 567/603 claims from single-point trains → zero-width bands (non-degenerate subset: 0.972). A width floor for train_n = 1 is noted for pre-registration *before* the next sweep, not patched retroactively.
6. **Same-polity historical bridge lane: n = 3** — Wilson [0.061, 0.792] is consistent with anything; no width model fixes a sample-size problem.
7. **Event tape has no empire-level polities** — `shock_windows_for_polity("ottoman_empire")` returns no windows; the mechanism is live but the tape names successor states only.

9. The load-bearing methodological commitments

Ranked by how much they changed conclusions today:

1. **Proper scoring rule as primary metric** — coverage alone would have inverted the shock finding (§6.4) and rewarded the climatology floor.
2. **Controls travel with every discovery loop** — the shuffle control (§3) and permutation control (§6.5) each caught an over-claim before it shipped.
3. **Selection/confirmation separation with leakage checks** — the cross-cut leak (§6.2) was found *before* selection ran; the amendment is documented in the plan changelog.
4. **Settlement-only learning with graded weights** — the ledger's trust numbers moved in interpretable, defensible directions (§2.3) rather than by fiat.
5. **Estimating parameters from data rather than tuning on outcomes** — the nugget (§7.2) was measured from cross-source spreads; the confirmatory tape stayed unburned.
6. **Abstention as a free, first-class outcome** — everywhere (AR1, LLM, analog, conservation) silence carried information instead of being forced into noise.

10. Open items, in leverage order

1. **E8 mid-century ingest (1930–1990)** — one ingest unblocks four consumers: co-variance strict tier, conservation claims, shock-split calm side, same-polity bridge holdouts.
2. **Empire-membership mapping on the event tape** — activates shock widening for Ottoman-era series.
3. **train_n = 1 width floor** — pre-register, then re-sweep.
4. **Analog candidate at larger n** — paired CI resolution needs the denser desk.
5. **Definitional lanes in the forecaster** — per-target-source blocks exist; a CJP↔affiliation discount at claim time is the next step.

11. Reproducibility

Result	Artifact	Runner
Trust + routing	`PHASE2B_TRUST.json`	`scripts/run_phase2b_connascence.py`
Co-variance + controls	`PHASE2B_CONNASCENCE.json`	same
LLM proposer	`PHASE2B_LLM_ENRICHMENT.json`	`scripts/run_llm_enrichment.py --full`
Pre-registered backtest	`PHASE3_BACKTEST.json`	`scripts/run_phase3_backtest.py`
E1–E7	`PHASE3_CUT_SWEEP / WIDTH_ABLATION / CANDIDATES / BRIDGE_CURVES .json`	`scripts/run_phase3_experiments.py`
Bridge v2	`PHASE3_BRIDGE_V2.json`, `BRIDGE_V2_BACKFILL.jsonl`	`scripts/run_bridge_v2.py`

Contracts: `make test` (235), phase-scoped `make test-phase{1,2,3}`.